SET-N

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: Max Marks: 30

Instructions:

I) Attempt any **one** question from each section.

II) Each question carries **10 marks.**

III) The **Journal** and **Viva-Voce** carry **5 marks.**

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to sort array element in ascending order using bubble sort method.

Q.2 Write a program to implement stack with its basic operations.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table employee by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Data Type	Constraint
Emp_ID	INT	Primary Key
Emp_Name	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	—
Designation	VARCHAR(30)	—
Salary	DECIMAL(10,2)	—
City	VARCHAR(30)	—
Joining_Date	DATE	—

a) Display all employee details.

b) Display employees working in the Computer Science department.

c) Display employees earning a salary greater than 40,000.

d) Display employees whose salary is between 25,000 and 50,000.

e) Display employees whose names end with 'a'.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Student:

Field Name	Data Type	Constraint
Student_ID	INT	Primary Key
Student_Name	VARCHAR(50)	NOT NULL
Class	VARCHAR(20)	—
City	VARCHAR(30)	—

B] Book_Issue:

Field Name	Data Type	Constraint
Issue_ID	INT	Primary Key
Student_ID	INT	Foreign Key referencing Student(Student_ID)
Book_Name	VARCHAR(100)	NOT NULL
Issue_Date	DATE	—
Return_Date	DATE	—
Fine	DECIMAL(7,2)	—

a) Display student names with the names of books issued using an `INNER JOIN`.

b) Display all students and their issued books using a `LEFT JOIN`.

c) Display students who have issued more than one book.

d) Display students who have not issued any books using a subquery.

e) Display students who have incurred a fine greater than ₹100.